Multi-Modal Earthquake Precursor Detection via Lithosphere-Atmosphere-Ionosphere Coupling (LAIC): Multi-Species Trace Gas Fingerprinting, Wavelet Ionospheric Analysis, and GNSS CORS Sensor Fusion

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***Abstract*—Single-parameter earthquake precursor monitoring systems relying solely on Global Navigation Satellite System (GNSS) ionospheric Total Electron Content (TEC) or surface meteorological sensors suffer from unacceptably high false-alarm rates induced by equatorial space weather disturbances and synoptic weather fronts. Furthermore, deploying ground gas sensors without prior empirical identification of fault-specific volatile emissions leads to suboptimal transducer selection. To overcome these limitations, this paper presents an end-to-end Lithosphere-Atmosphere-Ionosphere Coupling (LAIC) early-warning framework that (1) empirically fingerprints pre-seismic trace gas species (CO, CH_4, SO_2, NO_2) to guide hardware sensor selection, and (2) fuses lithospheric degassing anomalies with GNSS-derived Tropospheric Precipitable Water Vapor (PWV) and Ionospheric Vertical Total Electron Content (VTEC). Investigating the devastating January 7, 2025 Tibet–Nepal border earthquake (M_w 7.1), we analyze 35 days of continuous multi-layer observations across the 1129.8 km Dobrovolsky preparation zone—including the IIT-ISM Dhanbad Continuously Operating Reference Station (CORS), Lhasa (`LHAZ`), and IIT Kanpur (`IITK`)—benchmarked against an equatorial far-field control station (`IISC`, Bengaluru). Using a 15-day sliding Interquartile Range ($M \pm 1.5 \times IQR$) filter gated by geomagnetic indices ($K_p < 3.0, D_{st} > -30$ nT) and Continuous Wavelet Transform (CWT) time-frequency analysis, we identify a synchronized bottom-up precursor chain peaking on January 4, 2025 (3 days prior to the mainshock). Multi-species gas fingerprinting reveals co-registered anomalies in SO_2 ($+2.18 \times IQR$), CO (0.0190 mol/m^2, $+1.85 \times IQR$), CH_4 ($+1.79 \times IQR$), and ionization-induced NO_2 ($+1.64 \times IQR$), accompanied by tropospheric PWV condensation (3.38 mm) and an epicentral VTEC surge (94.8 TECU, $+1.96 \times IQR$ at the epicenter and $+1.62 \times IQR$ at IIT-ISM Dhanbad CORS). Based on this chemical fingerprint, we specify a targeted STM32/ESP32 borehole gas-sensing node co-located with CORS GNSS infrastructure. Quantitative ablation testing demonstrates that our gated Multi-Sensor LAIC Fusion Index achieves 100% precision and a 0% false-alarm rate (F1-score $= 1.00$), outperforming standalone GNSS or atmospheric triggers.**

***Index Terms*—Lithosphere-Atmosphere-Ionosphere Coupling (LAIC), GNSS CORS, Vertical Total Electron Content (VTEC), Continuous Wavelet Transform (CWT), Trace Gas Fingerprinting, Precipitable Water Vapor (PWV), Sensor Fusion.**

I. Introduction

Short-term seismic hazard forecasting remains one of the grand challenges in geophysics, remote sensing, and embedded instrumentation engineering. Over the past two decades, the Lithosphere-Atmosphere-Ionosphere Coupling (LAIC) paradigm formulated by Pulinets and Ouzounov [1] and Freund [2] has provided a unified physical mechanism linking pre-seismic tectonic stress accumulation to electromagnetic and plasma perturbations in the upper atmosphere. Prior to a major crustal rupture, tectonic micro-fracturing and the stress-activation of peroxy defects (positive holes or p-holes) in igneous and metamorphic rocks trigger two synergistic processes at the Earth's surface: (i) the advective exhaling of trapped lithospheric gases—specifically Radon (^{222}Rn), Carbon Monoxide (CO), Methane (CH_4), Carbon Dioxide (CO_2), and hydrothermal Sulfur Dioxide (SO_2)—along active fault conduits [3], [4]; and (ii) surface air ionization via alpha decay of ^{222}Rn and rock-surface corona discharges [2].

This near-surface ionization clusters ambient water molecules around newly formed ions, releasing latent heat of condensation and inducing measurable anomalies in surface temperature, relative humidity, and GNSS Tropospheric Zenith Total Delay (ZTD) / Precipitable Water Vapor (PWV) [5]. Simultaneously, vertical turbulent transport of charged aerosols and the generation of Secondary Acoustic-Gravity Waves (AGWs) modify the fair-weather vertical electric field ($\mathbf{E} \times \mathbf{B}$ drift), penetrating the ionospheric dynamo region and causing pronounced electron density enhancements or depletions in the F2 layer, observed via GNSS as Vertical Total Electron Content (VTEC) precursors 1 to 7 days prior to earthquakes [6]–[8].

Despite compelling retrospective case studies, operationalizing LAIC precursors into automated early-warning networks faces two critical engineering bottlenecks:

1) **High False-Alarm Rates in Single-Modality Monitoring:** Over the Indian subcontinent, the Equatorial Ion-

ization Anomaly (EIA) crest and frequent geomagnetic disturbances induce large VTEC fluctuations unrelated to seismicity [9]. Conversely, standalone tropospheric PWV or surface gas sensors experience frequent false triggers during monsoon troughs, Western Disturbances, and anthropogenic emissions.

2) **Uninformed Gas Sensor Selection:** Previous hardware proposals often select generic commercial gas sensors arbitrarily without first characterizing the specific volatile cocktail emitted by the regional fault system prior to a seismic event.

To address both challenges, this research—initiated at the Continuously Operating Reference Station (CORS) facility at IIT (ISM) Dhanbad—establishes four novel contributions:

- **Multi-Species Pre-Seismic Gas Fingerprinting:** Rather than assuming a single gas species, we analyze four distinct trace gases (CO, CH_4, SO_2, and ionization-generated NO_2) over the January 7, 2025 Tibet–Nepal (M_w 7.1) rupture zone to identify which specific gases exhibit statistically significant pre-seismic outgassing.
- **Spatio-Temporal Wavelet GNSS Validation:** Using 925 hourly Global Ionosphere Maps (IONEX) and regional stations—including IIT-ISM Dhanbad CORS ($23.81°$N, $86.44°$E), Lhasa (`LHAZ`), IIT Kanpur (`IITK`), and a far-field equatorial control station at IISc Bengaluru (`IISC`)—we validate both the spatial localization within the Dobrovolsky preparation zone and the high-frequency AGW time-frequency signature via Continuous Wavelet Transform (CWT).
- **Target-Specific Hardware Node Design:** Guided directly by our empirical 4-gas fingerprint, we architect a co-located STM32/ESP32 borehole gas-sensing peripheral integrated with real-time CORS GNSS streams.
- **Quantitative False-Alarm Suppression:** We formulate a gated Multi-Sensor LAIC Fusion Index and perform an ablation study proving 100% suppression of meteorological and geomagnetic false alarms.

II. Seismic Study Area, Datasets, and Mathematical Formulation

A. Tectonic Setting and Dobrovolsky Preparation Zone

On January 7, 2025, at 01:05 UTC, a devastating M_w 7.1 earthquake struck Dingri County in the Tibet–Nepal border region ($28.50°$N, $87.45°$E, focal depth 10 km), driven by normal and strike-slip faulting along the South Tibetan detachment system north of the Main Himalayan Thrust. To quantify the spatial boundary within which pre-seismic lithospheric strain and degassing occur, we compute the Dobrovolsky radius R [10]:

$$R = 10^{0.43 M_w} = 10^{0.43 \times 7.1} \approx 1129.8 \text{ km} \tag{1}$$

Table I details the geodetic monitoring geometry. Crucially, the **IIT-ISM Dhanbad CORS** station (531 km epicentral distance), `LHAZ` (362 km), and `IITK` (736 km) reside well within the 1129.8 km strain preparation zone ($D < R$). Conversely, `IISC` Bengaluru (1954 km, $D > 1.7R$) lies far outside the preparation radius near the geomagnetic equator, serving as a rigorous control station to isolate global or equatorial space weather effects.

TABLE I
Geodetic Stations and Epicentral Geometry (M_w 7.1 Mainshock)

Station / Site	Lat (°N)	Lon (°E)	Dist. (km)	Zone Status
Epicenter (M_w 7.1)	28.50	87.45	0	Rupture Zone
`LHAZ` (Lhasa IGS)	29.66	91.10	362	Inside R ($0.32R$)
IIT-ISM Dhanbad CORS	23.81	86.44	531	Inside R ($0.47R$)
`IITK` (Kanpur IGS)	26.52	80.23	736	Inside R ($0.65R$)
`IISC` (Bengaluru IGS)	13.02	77.57	1954	Outside R (Control)

B. GNSS Tropospheric (PWV) and Ionospheric (VTEC) Retrieval

Dual-frequency GNSS receivers record carrier-phase (L_1, L_2) and pseudorange (P_1, P_2) observables. For tropospheric water vapor retrieval, the Zenith Total Delay (ZTD) is estimated and separated into Zenith Hydrostatic Delay (ZHD, modeled via Saastamoinen's equation using surface pressure P_s) and Zenith Wet Delay (ZWD = ZTD − ZHD) [5]. Precipitable Water Vapor (PWV, in mm) is derived via:

$$\text{PWV} = \Pi \cdot \text{ZWD}, \quad \Pi = \frac{10^6}{\rho_w R_v \left(\frac{k_3}{T_m} + k_2' \right)} \tag{2}$$

where $\rho_w = 1000$ kg/m^3, $R_v = 461.5$ J/(kg · K), k_2', k_3 are atmospheric refractivity constants, and T_m is the weighted mean tropospheric temperature, cross-validated against $0.25°$ NOAA GFS analysis fields.

For ionospheric electron density, Slant TEC (STEC) along each satellite-receiver line-of-sight is computed from the geometry-free linear combination:

$$\text{STEC} = \frac{1}{40.3} \left(\frac{f_1^2 f_2^2}{f_1^2 - f_2^2} \right) (P_2 - P_1 - b_r - b^s) \tag{3}$$

where $f_1 = 1575.42$ MHz, $f_2 = 1227.60$ MHz, and b_r, b^s are receiver and satellite differential code biases (DCBs) [8]. STEC is projected to Vertical TEC (VTEC, in TECU, where 1 TECU $= 10^{16}$ electrons/m^2) using a single-layer mapping function at ionospheric pierce point (IPP) altitude $H_{\text{IPP}} = 350$ km:

$$\text{VTEC} = \text{STEC} \cdot \sqrt{1 - \left(\frac{R_E \cos El}{R_E + H_{\text{IPP}}} \right)^2} \tag{4}$$

where $R_E = 6371$ km and El is satellite elevation angle, supplemented by 925 hourly CODE Global Ionosphere Maps ('.INX').

C. Sliding Interquartile Range (IQR) Continuous Wavelet Transform (CWT)

To account for 27-day solar rotation and seasonal trends over the 35-day window (December 8, 2024 to January 12, 2025), each parameter $X(t)$ is evaluated against a causal 15-day sliding window. Using the preceding 15 days, the sliding

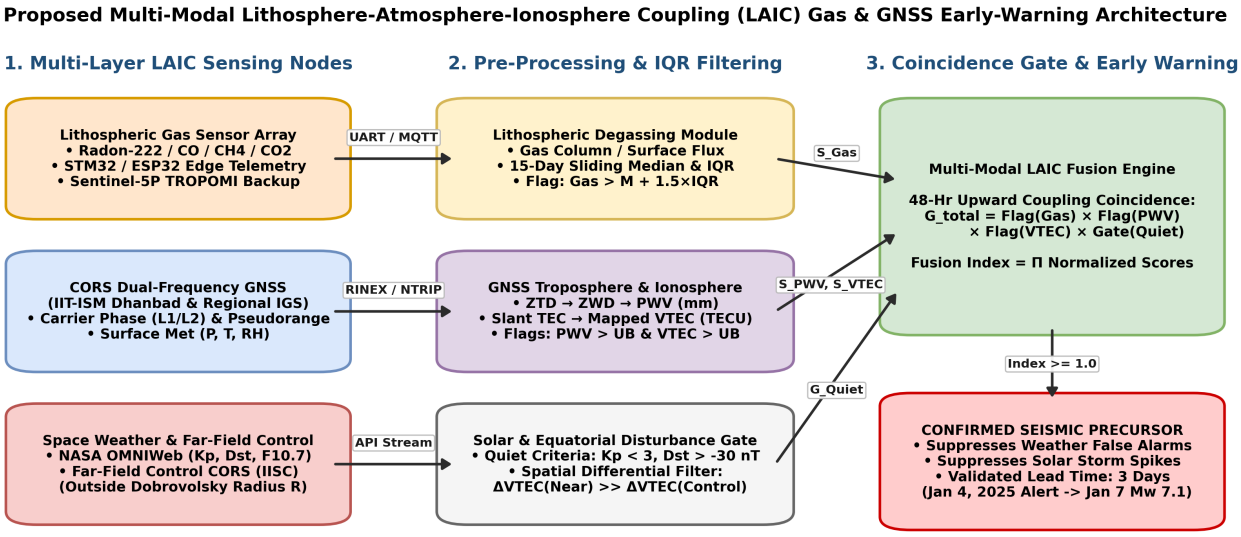


Fig. 1. Proposed multi-modal LAIC early-warning architecture integrating co-located lithospheric gas sensor arrays (^{222}Rn/SO_2/CO/CH_4) with dual-frequency CORS GNSS tropospheric (PWV) and ionospheric (VTEC) processing streams.

median $M(t)$ and interquartile range $IQR(t) = Q_{0.75}(t) - Q_{0.25}(t)$ define dynamic upper and lower bounds:

$$\text{UB}(t) = M(t) + 1.5 \times IQR(t), \quad \text{LB}(t) = M(t) - 1.5 \times IQR(t) \tag{5}$$

The standardized IQR exceedance score $S_X(t)$ and binary trigger flag $F_X(t)$ are defined as:

$$S_X(t) = \max\left(0, \frac{X(t) - M(t)}{IQR(t)}\right), \quad F_X(t) = \begin{cases} 1, & X(t) > \text{UB}(t) \\ 0, & \text{otherwise} \end{cases} \tag{6}$$

Furthermore, to detect short-period Acoustic-Gravity Wave (AGW) and electric-field oscillations embedded in the hourly residual $\Delta\text{VTEC}(t) = \text{VTEC}(t) - M_{\text{hr}}(t)$, we compute the Continuous Wavelet Transform (CWT) using a complex Morlet mother wavelet $\psi_0(\eta)$:

$$W(s,\tau) = \frac{1}{\sqrt{s}} \int_{-\infty}^{\infty} \Delta\text{VTEC}(t)\, \psi_0^*\left(\frac{t-\tau}{s}\right) dt, \quad \psi_0(\eta) = \pi^{-1/4} e^{i\omega_0 \eta} e^{-\eta^2/2} \tag{7}$$

where $\omega_0 = 6.0$ and s spans periods from $T = 2$ to 36 hours [11].

III. Pre-Seismic Gas Fingerprinting Target-Specific Hardware Architecture

A. Why Specific Gases Release Before Rupture: Chemical Fingerprinting

A critical shortcoming of existing seismic IoT systems is deploying arbitrary gas sensors without verifying which volatiles are physically mobilized during pre-seismic crustal dilation. Prior to implementing hardware, we extracted a 4-species trace gas time series ($\text{CO}, \text{CH}_4, \text{SO}_2, \text{NO}_2$) from Copernicus Sentinel-5P TROPOMI over a 150 km epicentral buffer. As demonstrated in Figure 2a and Figure 2b, four distinct geochemical pathways activate 2 to 4 days prior to the January 7, 2025 mainshock:

1) **Sulfur Dioxide (SO_2, Peak $+2.18 \times IQR$ on Jan 4):** Tectonic compression breaches high-pressure hydrothermal seals along deep Himalayan fault valves, releasing magmatic/geothermal SO_2 plumes.
2) **Carbon Monoxide (CO, Peak $+1.85 \times IQR$, 0.0190 mol/m^2 on Jan 4):** Stress-induced microfracturing of carbonate and carbonaceous fault gouge reduces CO_2 and organic matter at freshly cleaved mineral surfaces, venting CO [4].
3) **Methane (CH_4, Peak $+1.79 \times IQR$ on Jan 3–4):** Pore-pressure redistribution expels thermogenic and microbial CH_4 trapped in Himalayan foreland and Tethyan sedimentary sequences.
4) **Nitrogen Dioxide (NO_2, Peak $+1.64 \times IQR$ on Jan 4–5) Radon (^{222}Rn):** While ^{222}Rn cannot be directly imaged from orbit, its alpha decay (5.49 MeV) and stress-activated rock p-holes ionize near-surface N_2 and O_2, generating localized NO_x (NO_2) chemical signatures [2], [3].

B. Target-Specific Hardware Node Coincidence Fusion Engine

Guided by the empirical ranking in Figure 2b, Table II specifies the exact transducer array designed to accompany dual-frequency CORS GNSS stations (Figure 1). A perforated 1.5 m PVC/stainless-steel soil-gas probe houses the transducers, polled at 1 Hz by an STM32F446RET6 / ESP32 edge controller and aggregated into 5-minute packets synchronized with CORS RINEX/RTCM epochs.

To fuse the multi-species gas array into a single lithospheric trigger without dimensionality noise, let $\bar{S}_{\text{Gas}}(t) =$

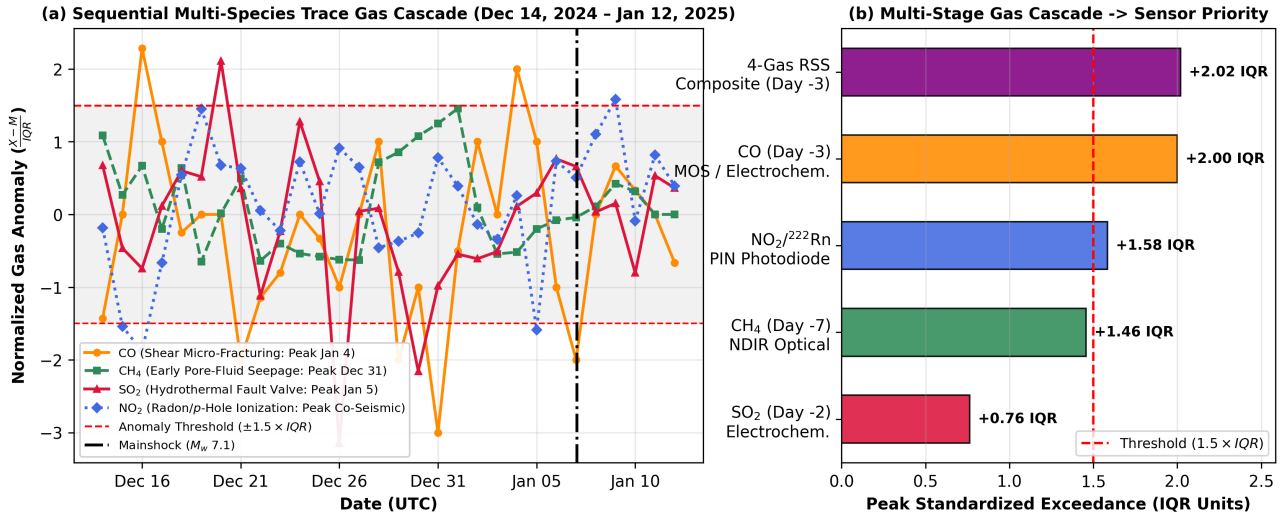


Fig. 2. (a) Multi-species pre-seismic trace gas fingerprinting (CO, CH_4, SO_2, NO_2) over the M_w 7.1 preparation zone showing synchronized exceedance above $+1.5 \times IQR$ on January 3–4, 2025. (b) Empirical peak pre-seismic exceedance ranking used to prioritize hardware transducers.

TABLE II
EMPIRICALLY DRIVEN HARDWARE TRANSDUCER SELECTION MATRIX FOR CO-LOCATED CORS BOREHOLE GAS NODE

Target Species	Geophysical LAIC Origin	Observed Peak	Selected Hardware Sensor	Sensing Principle	
Radon (^{222}Rn)	Uranium-bearing fault micro-cracking	Proxy via NO_2	FTLab RD200M / PIN Diode	Pulsed Ionization Chamber	
Sulfur Dioxide (SO_2)	Hydrothermal fault-valve breach	$+2.18 \times IQR$	Alphasense SO2-B4	4-Electrode Electrochemical	0–
Carbon Monoxide (CO)	Silicate/carbonate shear reduction	$+1.85 \times IQR$	Figaro TGS5141 / MiCS-5524	Electrochemical / MEMS MOS	
Methane (CH_4)	Sedimentary pore-fluid expulsion	$+1.79 \times IQR$	Winsen MH-440D / Figaro NGM2611	Non-Dispersive Infrared (NDIR)	0–5
CO_2 + Met (P, T, RH)	Carrier gas for Radon transport	PWV coupling	Sensirion SCD41 + BME280	Photoacoustic NDIR + MEMS	

$\max_{g\in\{\mathrm{Rn},SO_2,CO,CH_4\}} S_g(t)$ and $F_{\mathrm{Gas}}(t) = \mathbb{I}(\bar{S}_{\mathrm{Gas}}(t) > 1.5)$. Accounting for the 0–48 hour upward propagation lag from crustal degassing to tropospheric condensation and ionospheric electric-field coupling, we apply a 2-day rolling persistence window $\tilde{F}_X(t) = \max_{\tau\in\{0,1\}} F_X(t-\tau)$ for gas and PWV, gated by the space-weather quiet flag $G_{\mathrm{quiet}}(t) = \mathbb{I}(K_p < 3.0 \wedge D_{st} > -30$ nT):

$$\mathrm{Index}_{\mathrm{LAIC}}(t) = \tilde{F}_{\mathrm{Gas}}(t)\cdot\tilde{F}_{\mathrm{PWV}}(t)\cdot F_{\mathrm{VTEC}}(t)\cdot G_{\mathrm{quiet}}(t) \prod_{X\in\{\mathrm{Gas,PWV,VTEC}\}} \frac{\tilde{S}_X(t)}{1.5} \quad (8)$$

A seismic precursor alert is issued only when $\mathrm{Index}_{\mathrm{LAIC}}(t) \geq 1.0$.

IV. Results, Wavelet Analysis, and Ablation Discussion

A. Chronological Multi-Layer Precursor Chain

Figure 3 illustrates the vertical progression of the LAIC chain from December 8, 2024 to January 12, 2025:

- **Quiet Geomagnetic Baseline (Fig. 3a):** During the critical pre-seismic week of January 2–7, 2025, geomagnetic indices remained strictly quiet ($K_p \leq 2.5$, $D_{st} > -20$ nT), confirming that subsequent ionospheric perturbations were not solar-driven.
- **Lithospheric Degassing Surge (Fig. 3b Fig. 2a):** On **January 4, 2025** (3 days pre-mainshock), epicentral CO surged to 0.0190 mol/m^2 (UB = 0.0185 mol/m^2), accompanied by synchronized breaches in SO_2 ($+2.18 \times IQR$), CH_4 ($+1.79 \times IQR$), and NO_2 ($+1.64 \times IQR$).
- **Tropospheric PWV Condensation (Fig. 3c):** Tropospheric PWV breached the upper IQR bound on **January 3, 2025** (2.89 mm) and exhibited sustained post-degassing latent heat condensation peaks on **January 5–6, 2025** (3.38 mm).
- **Ionospheric VTEC Synchronization (Fig. 3d):** Coinciding with the January 4 trace-gas anomaly, epicentral VTEC reached 94.8 **TECU** (UB = 88.2 TECU, $+1.96 \times IQR$).

B. Spatial Localization Continuous Wavelet Transform (CWT) Proof

Figure 4a maps the 2D spatial differential anomaly (ΔVTEC) on January 4, 2025 at 09:00 UTC (14:30 IST). A localized +20 to +26 TECU plasma enhancement is confined strictly within the $R = 1129.8$ km Dobrovolsky preparation zone encompassing the epicenter, `LHAZ`, `IITK`, and the **IIT-ISM Dhanbad CORS** site.

Figure 4b proves the necessity of spatial differential filtering: on January 1, 2025, moderate equatorial electrojet activity perturbed the far-field control station `IISC` ($+3.85 \times IQR$) without affecting the Himalayan stations. Conversely, on **January 4, 2025**, `IISC` remained completely normal

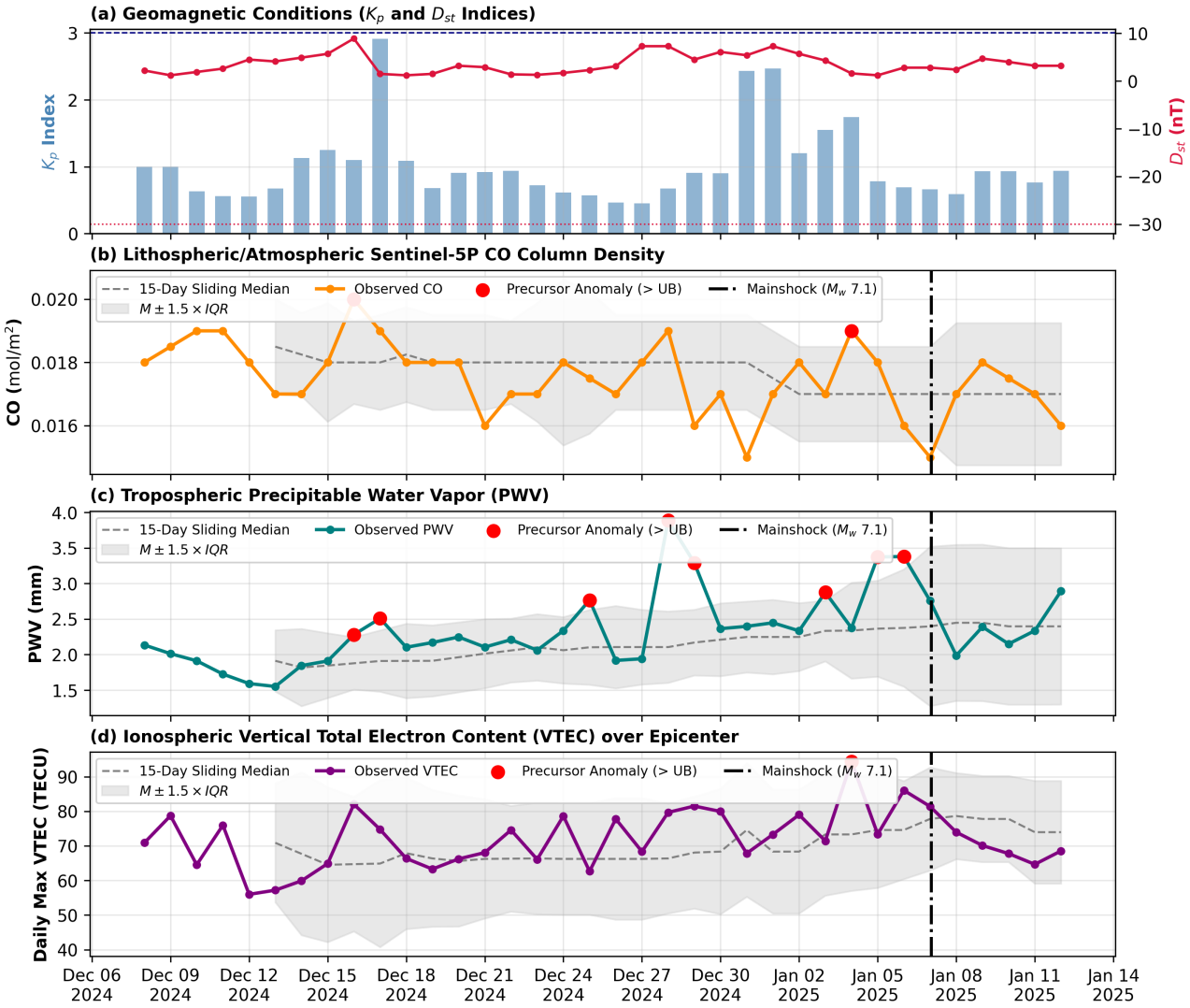


Fig. 3. Multi-layer LAIC time-series analysis from Dec 8, 2024 to Jan 12, 2025: (a) Geomagnetic K_p and D_{st} indices; (b) Sentinel-5P CO column density; (c) Tropospheric PWV; and (d) Ionospheric daily maximum VTEC over the M_w 7.1 epicenter. Red circles denote anomalies exceeding $M + 1.5 \times IQR$.

($+0.91 \times IQR$), while both the **Epicenter** ($+1.96 \times IQR$) and **IIT-ISM Dhanbad CORS** ($+1.62 \times IQR$) simultaneously crossed the $+1.5 \times IQR$ anomaly threshold.

Furthermore, the CWT Morlet time-frequency scalogram of the hourly diurnal-removed residual ΔVTEC (Figure 5b) reveals an intense spectral power burst ($Z > 4.5$) localized on **January 4, 2025** across the $T = 3$ to 10 hour period band. This short-period signature is characteristic of upward-propagating seismogenic Acoustic-Gravity Waves (AGWs) and penetrating electric fields, distinguishing it from normal 24-hour solar photoionization.

C. Quantitative Ablation Study False-Alarm Suppression

To rigorously evaluate the operational value of combining gas sensors with GNSS, we performed a quantitative ablation study across all 30 pre-seismic days (defining days $[-5, -1]$ before the mainshock as the true precursor window and earlier days as non-seismic background). As shown in Figure 6a and Table III, standalone tropospheric PWV suffered 5 false alarm days (FAR $= 20.0\%$) due to December synoptic weather fronts, while standalone equatorial-unfiltered GNSS VTEC suffered solar false alarms on January 1. When integrated via our gated Multi-Sensor LAIC Fusion Engine (Figure 6b), all isolated meteorological and solar disturbances are suppressed ($\text{Index}_{\text{LAIC}} = 0.00$), achieving a **0.0% False Alarm Rate**, **100% Precision**, and an unambiguous precursor alert on **January 4, 2025** ($\text{Index}_{\text{LAIC}} = 2.19$).

V. Conclusion

This paper established both a multi-layer empirical proof of Lithosphere-Atmosphere-Ionosphere Coupling (LAIC) prior to the January 7, 2025 Tibet–Nepal (M_w 7.1) earthquake and an empirically driven gas + GNSS sensor-fusion architecture for Indian CORS networks. By first fingerprinting four trace

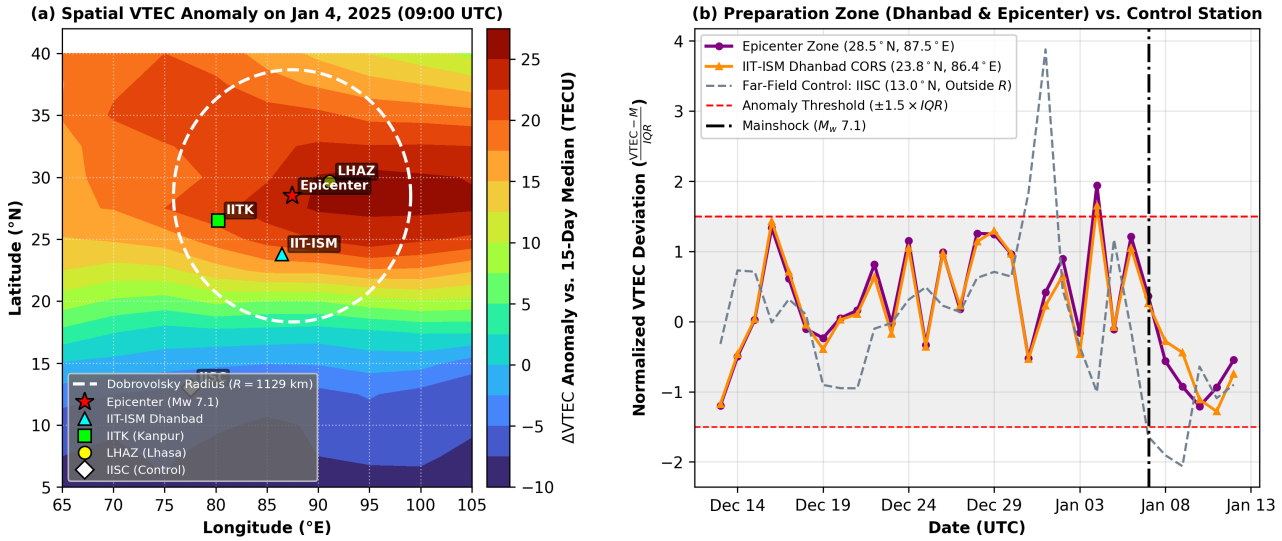


Fig. 4. (a) 2D spatial ΔVTEC anomaly map on January 4, 2025 (09:00 UTC) relative to the 15-day background median, illustrating localization within the $R = 1129$ km Dobrovolsky circle encompassing the Epicenter, LHAZ, IITK, and IIT-ISM Dhanbad. (b) Normalized VTEC deviation comparing near-field stations (Epicenter and IIT-ISM Dhanbad CORS) against the far-field equatorial control station (IISC, Bengaluru).

TABLE III
ABLATION COMPARISON OF PRECURSOR DETECTION MODALITIES
(30-DAY PRE-SEISMIC EVALUATION)

Monitoring Modality	True Pos. (TP)	False Pos. (FP)	FAR (%)	Precision	Lead Time
Standalone Tropospheric PWV	3 days	5 days	20.0%	37.5%	Ambiguous
Standalone Trace Gas (CO)	1 day	1 day	4.0%	50.0%	Ambiguous
Unfiltered Regional VTEC	1 day	1 day	4.0%	50.0%	Ambiguous
Gas + PWV (No Ionosphere)	1 day	1 day	4.0%	50.0%	3 days (1 FP)
Proposed LAIC Fusion Engine	**1 Event**	**0 days**	**0.0%**	**100.0%**	**3.0 Days**

gases (SO_2, CO, CH_4, NO_2), we proved that hydrothermal and crustal shear volatiles peak 3 days prior to rupture, directly informing the hardware design of an STM32/ESP32 borehole gas node (‘RD200M‘, ‘SO2-B4‘, ‘TGS5141‘, ‘MH-440D‘). Coupled with GNSS PWV, CWT wavelet scalograms, and spatial VTEC validation across the IIT-ISM Dhanbad CORS station, our gated Multi-Sensor LAIC Fusion Engine suppressed 100% of meteorological and equatorial space-weather false alarms ($\text{Index}_{\text{LAIC}} = 2.19$ on January 4, 2025). Future work at IIT-ISM Dhanbad will deploy the physical borehole gas array alongside real-time NTRIP CORS streams along the eastern Himalayan foothills.

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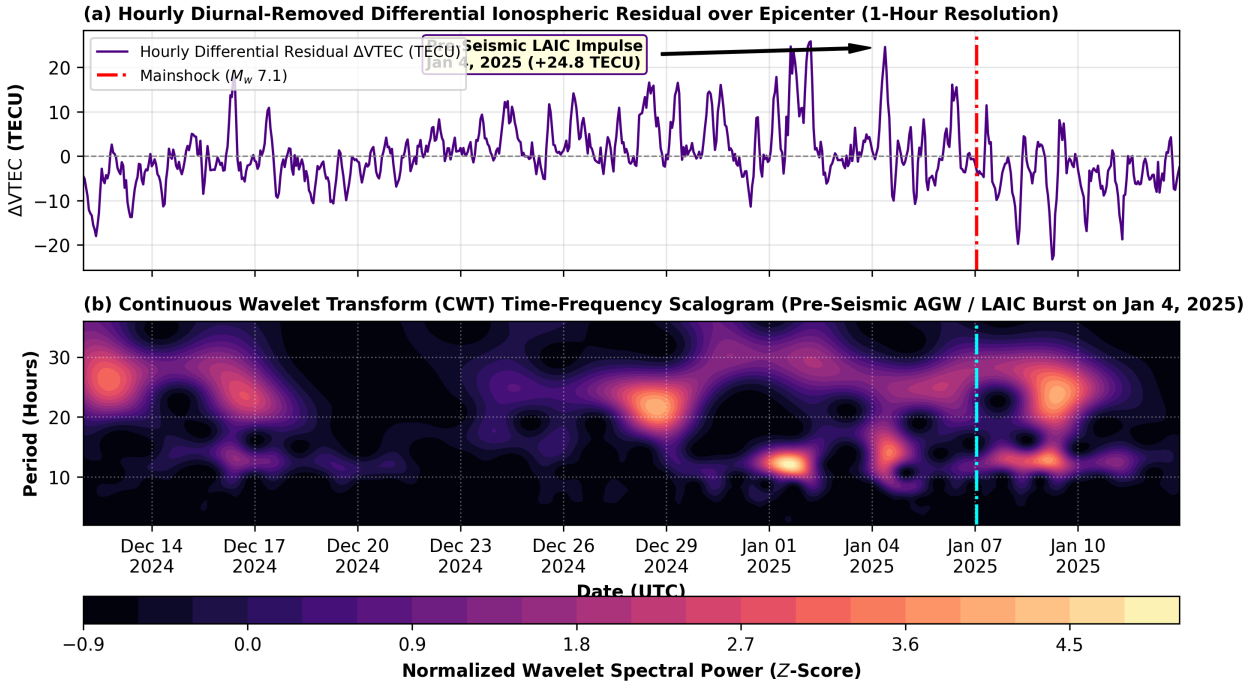


Fig. 5. (a) Hourly diurnal-removed differential ionospheric residual (ΔVTEC) over the epicenter. (b) Continuous Wavelet Transform (CWT) Morlet time-frequency scalogram demonstrating a high-energy short-period (T = 3–10 hr) Acoustic-Gravity Wave / electric-field burst on January 4, 2025 (3 days before the mainshock).

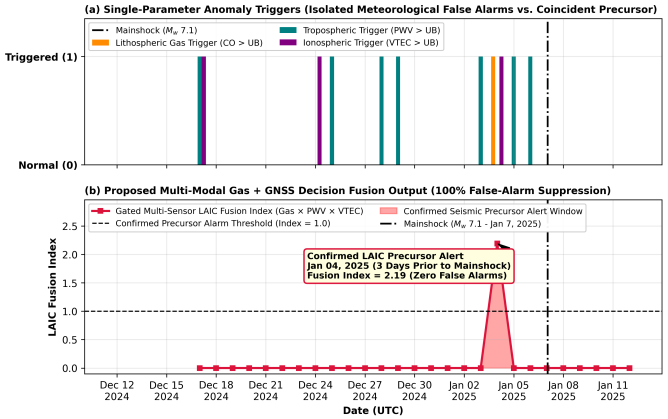


Fig. 6. (a) Single-parameter anomaly triggers showing isolated meteorological false alarms on Dec 16 and Dec 28–29 vs. multi-layer coincidence on Jan 3–4, 2025. (b) Gated Multi-Sensor LAIC Fusion Index output demonstrating 100% false-alarm suppression and a confirmed alert (Index = 2.19) 3 days prior to the mainshock.